

Davide Murari

POSTDOCTORAL RESEARCH ASSOCIATE

Department of Applied Mathematics and Theoretical Physics, University of Cambridge

✉ dm2011@cam.ac.uk | 🏠 <https://davidemurari.com/> | 📺 davidemurari | 🎓 Davide Murari

RESEARCH INTERESTS Geometric numerical methods, scientific machine learning, approximation theory, and data-driven discovery of dynamical systems.

Academic Positions and Affiliations

1 October 2024 – present Postdoctoral Research Associate in the Cambridge Image Analysis group of Professor Carola-Bibiane Schönlieb, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. Research within the EPSRC Programme Grant *The Mathematics of Deep Learning* (Maths4DL; EP/V026259/1) and the EPSRC-funded *ProbAI: A Hub for the Mathematical and Computational Foundations of Probabilistic AI* (EP/Y028783/1).

From 1 October 2026 Appointed Postdoctoral By-Fellow, Churchill College, University of Cambridge.

Education

- 2020–2024 PhD in Numerical Analysis, Norwegian University of Science and Technology (NTNU), supervised by Professors Elena Celledoni and Brynjulf Owren. Thesis: *Neural Networks, Differential Equations, and Structure Preservation*.
- 2018–2020 MSc in Mathematics (LM-40), University of Verona, Italy. Thesis: *Integrable Non-Hamiltonian Systems: From B-Integrability to Euler–Jacobi Theorem and Back*, supervised by Professor Nicola Sansonetto. Final grade: 110/110 cum laude.
- 2015–2018 BSc in Applied Mathematics (L-35), University of Verona, Italy. Thesis (English title): *An Overview of Ergodic Theory and Dynamical Billiards*, supervised by Professor Nicola Sansonetto. Final grade: 110/110 cum laude.

Publications

Author order in the publications below follows field-specific conventions and agreements among co-authors. Publications in which I played a leading role are marked with an asterisk (*).

- Forthcoming*** Ehrhardt, Matthias J., **Davide Murari**, and Ferdia Sherry. "Stable neural networks and connections to continuous dynamical systems." Forthcoming in *An Introduction to Scientific Machine Learning: Mathematical Foundations and Numerical Implementation*, edited by Victorita Dolean, Alexander Heinlein, and Olga Mula.
- 2026*** Canizares, Priscilla, **Davide Murari**, Carola-Bibiane Schönlieb, Ferdia Sherry, and Zakhar Shumaylov. "Symplectic Neural Flows for Modeling and Discovery." *SIAM Journal on Scientific Computing*, in press.
- 2026*** Furuya, Takashi, **Davide Murari** and Carola-Bibiane Schönlieb. "Approximation Theory for Lipschitz Continuous Transformers." In the *Forty-third International Conference on Machine Learning (ICML 2026)*, 2026.
- 2026** Liu, Chaoyu, **Davide Murari**, Lihao Liu, Yangming Li, Chris Budd, and Carola-Bibiane Schönlieb. "Enhancing Fourier Neural Operators with Local Spatial Features." *SIAM Journal on Scientific Computing* 48, no. 4 (2026): C991-C1010.
- 2026*** **Murari, Davide**, and Nicola Sansonetto. "From Euler–Jacobi to Bogoyavlenskij and back." *Nonlinearity* 39, no. 1 (2026): 015016.
- 2025*** **Murari, Davide**, Takashi Furuya, and Carola-Bibiane Schönlieb. "Approximation theory for 1-Lipschitz ResNets." In *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)*, 2025.
- 2025*** Celledoni, Elena, James Jackaman, **Davide Murari**, and Brynjulf Owren. "Predictions Based on Pixel Data: Insights from PDEs and Finite Differences." *Journal of Computational Physics* 538 (2025): 114166.
- 2025*** Celledoni, Elena, Ergys Çokaj, Andrea Leone, Sigrid Leyendecker, **Davide Murari**, Brynjulf Owren, Rodrigo T. Sato Martín de Almagro, and Martina Stavole. "Neural networks for the approximation of Euler's elastica." *Computer Methods in Applied Mechanics and Engineering* 435 (2025): 117584.

- 2024*** Eliasof, Moshe, **Davide Murari**, Ferdia Sherry, and Carola-Bibiane Schönlieb. "Resilient Graph Neural Networks: A Coupled Dynamical Systems Approach." ECAI - 27TH European Conference on Artificial Intelligence (2024): 1607-1614.
- 2024 Sherry, Ferdia, Elena Celledoni, Matthias J. Ehrhardt, **Davide Murari**, Brynjulf Owren, and Carola-Bibiane Schönlieb. "Designing stable neural networks using convex analysis and ODEs." Physica D: Nonlinear Phenomena 463 (2024): 134159.
- 2023*** Celledoni, Elena, **Davide Murari**, Brynjulf Owren, Carola-Bibiane Schönlieb, and Ferdia Sherry. "Dynamical Systems–Based Neural Networks." SIAM Journal on Scientific Computing 45, no. 6 (2023): A3071-A3094.
- 2023*** Celledoni, Elena, Andrea Leone, **Davide Murari**, and Brynjulf Owren. "Learning Hamiltonians of constrained mechanical systems." Journal of Computational and Applied Mathematics 417 (2023): 114608.
- 2022 Celledoni, Elena, Ergys Çokaj, Andrea Leone, **Davide Murari**, and Brynjulf Owren. "Lie group integrators for mechanical systems." International Journal of Computer Mathematics 99, no. 1 (2022): 58-88.

Preprints

An asterisk (*) indicates a leading role.

- 2026*** **Murari, Davide**, Erik Jansson, Chris Budd OBE, and Carola-Bibiane Schönlieb. "Learning piecewise-smooth dynamical systems." *arXiv preprint arXiv:2608.19785*, 2026.
- 2026*** **Murari, Davide**, Marta Ghirardelli, Ben Adcock, Elena Celledoni, Brynjulf Owren, and Carola-Bibiane Schönlieb. "1-Lipschitz Neural Networks on Hadamard Manifolds." *arXiv preprint arXiv:2607.19335*, 2026.
- 2025*** Massucco, Alex, **Davide Murari**, and Carola-Bibiane Schönlieb. "Deep Network Trainability via Persistent Subspace Orthogonality." *arXiv preprint arXiv:2508.02882* (2025).
- 2025 De Marinis, Arturo, **Davide Murari**, Elena Celledoni, Nicola Guglielmi, Brynjulf Owren, and Francesco Tudisco. "Approximation properties of neural ODEs." *arXiv preprint arXiv:2503.15696* (2025).
- 2024*** Betcke, Marta M., Lisa Maria Kreusser, and **Davide Murari**. "Parallel-in-Time Solutions with Random Projection Neural Networks." *arXiv preprint arXiv:2408.09756* (2024).

Teaching and supervision

- May 2026 Co-designed the new non-examinable course *Scientific Machine Learning* for the MPhil in Data Intensive Science, University of Cambridge, and delivered a one-hour lecture.
- 10–14 November 2025 Designed and delivered the three-lecture Master's-level mini-course *Neural Networks as Dynamical Systems*, University of Verona.
- 29 September – 3 October 2025 Invited lecturer at the 11th CISM–AIMETA Advanced Course on *Machine Learning for Solid Mechanics*, delivering three lectures (4.5 hours) on the mathematical foundations of deep learning, dynamical-systems-based neural networks, and scientific machine learning.
- July–September 2025 Independently supervised two undergraduate research interns, Moonis Haider and Claudia Franco, at the University of Cambridge through the Philippa Fawcett Internship and Summer Research in Maths (SRIM) programmes.
- February 2025 – December 2025 Co-supervised Andrea Meschieri's master's thesis at Politecnico di Milano, Light SDE Matching: A Data-Driven Framework for Learning Stochastic Differential Equations with Applications in Finance.
- January–March 2025 Designed and delivered the 16-hour non-examinable Part III course *Geometric Numerical Analysis and Deep Learning*, University of Cambridge, and wrote the accompanying lecture notes.
- November 2024 Designed and delivered the Master's-level mini-course *Geometric Methods for Differential Equations and Learning Applications*, University of Verona.
- 2021–2023 Teaching Assistant, *Numerical Solution of Partial Differential Equations Using Element Methods* (TMA4220), NTNU (Fall 2021, Fall 2022, Fall 2023).
- Spring 2022 Teaching Assistant, *Optimization 1* (TMA4180), NTNU.
- Spring 2021 Teaching Assistant, *Differential Equations and Dynamical Systems* (TMA4165), NTNU.

Academic Service

Cambridge Image Analysis	Deputy Head, Cambridge Image Analysis group, University of Cambridge. Supporting Professor Carola-Bibiane Schönlieb in the coordination of group activities, together with Ander Biguri and Michael Roberts. Responsibilities include organising the weekly group seminar, coordinating research visitors, and helping organise internal activities such as the group retreat.
Event organisation	- Co-organiser, SCML 2026: International Conference on Scientific Computing and Machine Learning, University of Bath, 14–17 September 2026. - Co-organiser, Mathematical Foundations of Deep Learning Summer School, University of Cambridge, 3–7 August 2026. A one-week summer school attended by 50 graduate students and early-career researchers, supported by Maths4DL and ProbAI.
Mini-symposium organisation	- SciCADE 2026, <i>Improving Numerical Methods with Machine Learning</i>, Edinburgh, 29 June–3 July 2026. - SciCADE 2024, <i>Dynamical Systems, Structure Preservation and Deep Learning</i>, Singapore, 15–19 July 2024. - Geometric Science of Information (GSI) 2023, <i>Deep Learning: Methods, Analysis and Applications to Mechanical Systems</i>, St. Malo, 30 August–1 September 2023.
Editorial work	Assisted Elena Celledoni in editing the <i>ECMI Annual Reports</i> 2021 and 2022, together with Andrea Leone.
Peer review	Referee for the <i>Journal of Machine Learning Research</i> , <i>Mathematics of Computation</i> , <i>SIAM Journal on Scientific Computing</i> , <i>SIAM Journal on Mathematics of Data Science</i> , <i>IMA Journal of Numerical Analysis</i> , <i>IMA Journal of Applied Mathematics</i> , <i>Journal of Computational Physics</i> , <i>Journal of Computational and Applied Mathematics</i> , and <i>Calcolo</i> .

External funding

July 2026	London Mathematical Society Early Career Researcher Travel Grant (£500), awarded to support participation and an invited presentation at FoCM 2026, Vienna.
January 2024	Funding from the Trond Mohn Foundation's <i>Pure Mathematics in Norway</i> programme (NOK 12,000), supporting participation in the Geilo Winter School <i>Graphs and Applications</i> .
March–May 2023	Simons Fellow, Isaac Newton Institute for Mathematical Sciences, with £1,400 of support during the thematic programme <i>The Mathematical and Statistical Foundation of Future Data-Driven Engineering</i> .
January 2023	Funding from the Trond Mohn Foundation's <i>Pure Mathematics in Norway</i> programme (NOK 10,000), supporting a research visit by Professor Nicola Sansonetto (University of Verona) to NTNU to make progress on our joint research on integrable dynamical systems.
October 2021	Isaac Newton Institute participation support (£330) for the thematic programme <i>Mathematics of Deep Learning</i> .

Research visits

March–May 2023	Visiting researcher, Isaac Newton Institute for Mathematical Sciences, Cambridge, during the thematic programme <i>The Mathematical and Statistical Foundation of Future Data-Driven Engineering</i> .
October 2021	Visiting researcher, Isaac Newton Institute for Mathematical Sciences, Cambridge, during the thematic programme <i>Mathematics of Deep Learning</i> .

Invited and conference talks

Slides for most talks are available at davidemurari.com/talks/.

September 2026	Seminar, Phi-ML meets Engineering, Alan Turing Institute (online). <i>Physics-informed discovery of non-smooth dynamical systems from trajectory data</i> .
September 2026	SCML 2026, University of Bath, UK. Contributed talk: <i>Discovering piecewise-smooth dynamics from trajectory data</i> .
July 2026	FoCM 2026, Vienna, Austria. Workshop talk: <i>Symplectic Neural Flows</i> .
June 2026	SciCADE 2026, Edinburgh, UK. Minisymposium talk: <i>Discovering Piecewise-Smooth Dynamics from Trajectory Data</i> .
May 2026	REMODEL Conference, Trondheim, Norway. Workshop talk: <i>Approximation Theory for 1-Lipschitz ResNets</i> .

- April 2026 EUROMECH Colloquium 662 *Physics-Enhanced Machine Learning and Data-Driven Nonlinear Dynamics*, Como, Italy. Workshop talk: *Discovering Piecewise-Smooth Dynamics from Trajectory Data*.
- October 2025 Seminar, Vector Institute, Toronto, Canada (online). *Approximation Theory for 1-Lipschitz ResNets*.
- July 2025 Third Joint SIAM/CAIMS Annual Meetings, Montréal, Canada. Minisymposium talk: *Symplectic Neural Flows*.
- April 2025 NOMADS Seminar, Gran Sasso Science Institute, L'Aquila, Italy. *Symplectic Neural Flows for Modelling and Discovery*.
- October 2024 Oberwolfach Workshop *Deep Learning for PDE-based Inverse Problems*, Germany. Workshop talk: *Dynamical Systems-Based Structured Networks*.
- October 2024 SIAM Conference on Mathematics of Data Science, Atlanta, USA. Minisymposium talk: *Structure-Preserving Solutions of Hamiltonian Systems Based on Neural Networks*.
- July 2024 SciCADE 2024, Singapore. Minisymposium talk: *Neural Network Aided Simulation of Ordinary Differential Equations*.
- February 2024 BIRS Workshop *Structured Machine Learning and Time-Stepping for Dynamical Systems*, Banff, Canada. Workshop talk: *Improving the Robustness of Graph Neural Networks with Coupled Dynamical Systems*.
- September 2023 TES Conference on Mathematical Optimization for Machine Learning, Berlin, Germany. Minisymposium talk: *Predictions Based on Pixel Data: Insights from PDEs and Finite Differences*.
- August 2023 ICIAM 2023, Tokyo, Japan. Minisymposium talk: *Structured Neural Networks and Some Applications*.
- June 2023 ECMI Conference 2023, Wrocław, Poland. Minisymposium talk: *Learning Hamiltonians of Constrained Mechanical Systems*.
- June 2023 FoCM 2023, Paris, France. Workshop talk: *Structured Neural Networks and Some Applications to Dynamical Systems*.
- May 2023 Isaac Newton Institute thematic programme *The Mathematical and Statistical Foundation of Future Data-Driven Engineering*, Cambridge, UK. Programme talk: *Neural Networks Motivated by Dynamical Systems*.
- February 2023 SIAM Conference on Computational Science and Engineering, Amsterdam, Netherlands. Minisymposium talk: *Structured Neural Networks and Their Relevance for Mechanical Systems*.
- December 2022 Workshop *Theoretical and Computational Aspects of Dynamical Systems (HB60)*, Trysil, Norway. Workshop talk: *Dynamical Systems-Based Neural Networks*.
- July 2022 SciCADE 2022, Reykjavík, Iceland. Minisymposium talk: *Structure-Preserving Neural Networks Coming from ODE Models*.
- June 2022 Workshop *Math Meets Industry*, Trondheim, Norway. Workshop talk: *Structured Neural Networks Motivated by Dynamical Systems*.
- May 2022 Machine Learning and Dynamical Systems Seminar, Alan Turing Institute (online). *Learning Hamiltonians of Constrained Mechanical Systems*.
- March 2022 MaGIC 2022, Ilsetra, Øyer, Norway. Workshop talk: *Robustness of Neural Networks for Classification Problems*.
- November 2021 Lorentz Center workshop *Computational Mathematics and Machine Learning*, Leiden, the Netherlands. Workshop talk: *Learning the Hamiltonian of Some Constrained Mechanical Systems*.
- September 2021 NUMDIFF-16, Halle, Germany. Minisymposium talk: *Learning the Hamiltonian of Some Classes of Mechanical Systems*.
- March 2021 MaGIC 2021, Ilsetra, Øyer, Norway. Workshop talk: *Lie Group Integrator's Approach to the N-fold Pendulum*.

Science communication

Founder of *Mathone* (2015–present), a public mathematics education and communication project spanning video and written content. The YouTube channel has approximately 8.9k subscribers and over 800k views, with current English-language content focused on applied mathematics, numerical analysis, and machine learning. I also write the Mathone Substack newsletter.